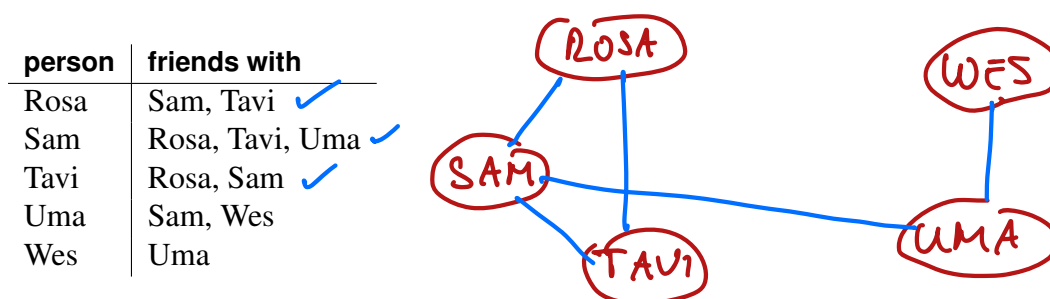


Goal: See **why** we might model a situation with a graph, and learn some vocabulary.

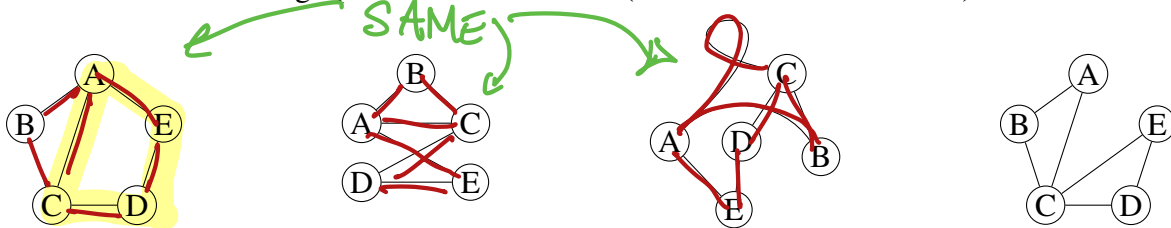
1. **vertex** (plural: **vertices**): *points that represent things*
edge: *connection between pairs of vertices (often a line)*

situation	vertices are...	edges are...
The road system connecting some towns in Alaska.	<i>towns</i>	<i>roads</i>
The streets in a neighborhood that a snowplow has to clear after a storm.	<i>intersections</i>	<i>roads</i>
A group of people, and who is friends with whom.	<i>people</i>	<i>if friends</i>
Classes offered at this university, and the students that take them.	<i>classes</i>	<i>students in a class</i>
	<i>students</i>	<i>in same class?</i>

2. We have information about five people and who is friends with whom. Draw a graph.



3. Are these all the same graph? How can we tell? (What does “same” mean?)



Edges?

AB BC
AC CD
AE DE

EDGES

AB BC
AC CD
AE DE

EDGES

AB BC
AC CD
AE DE

No quadrilateral!

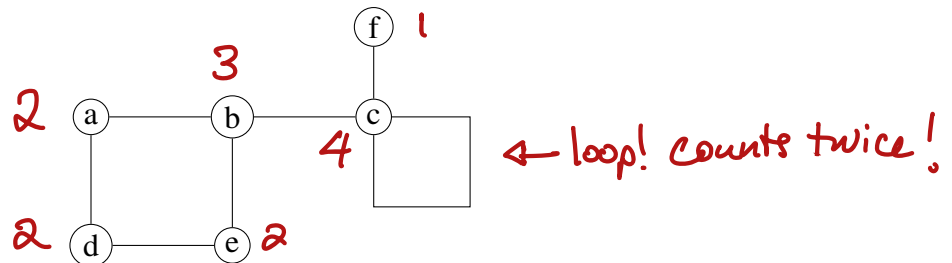
Edges

AB CE
AC CD
BC ED

* Same = Same # vtx, Same # edges,
Same connections (regardless of label)

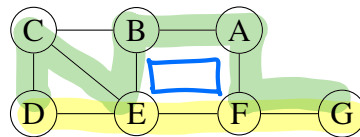
4. Below is a neighborhood: each vertex is an intersection, each edge is a street. How busy is each intersection?

Degree # edges at each vertex
Loop degree ← count in/out edge twice!



degree of a: 2 b: 3 c: 4 d: 2 e: 2 f: 1

5. Below is a road network: each vertex is a town, each edge is a road.



- (a) Can I get from here to there?

path: a sequence of vertices & edges

Find a path from D to G: DEFG or DCEBAFG

- (b) Can I leave home and get back?

circuit: path that starts and stops at the same vertex

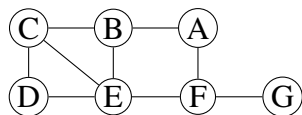
Find a circuit starting at E: EBAF

- (c) Is everything reachable at all?

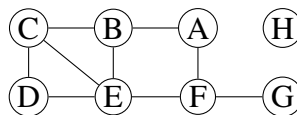
connected: there is a path between any pair of vertices

Not: You can FIND a pair w/no path between them

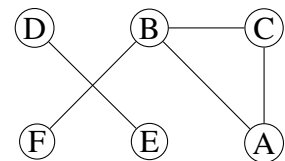
Which graphs are connected?



Yes!



No! Can't get to H



No! No path from B to E